MUHAMMAD ALI MUGHAL

AI Engineer

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PROFESSIONAL SUMMARY

Accomplished Artificial Intelligence Engineer with over 4 years of experience designing and implementing AI-driven solutions, specializing in predictive analytics, machine learning, generative AI, and process automation. Proven track record leading cross-functional teams to deliver complex AI and full-stack projects that drive business performance and operational efficiency. Skilled in building scalable cloud-based architectures, agentic AI workflows, and no-code/low-code automation platforms to democratize AI adoption. Seeking to leverage deep technical expertise and innovative problem-solving skills.

TECHNICAL SKILLS

- Data Science, Machine Learning, Python, Django, TensorFlow, Pandas, NumPy, OpenCV, Scikit-Learn
- AI, Deep Learning, Flask, GPT-4, Hugging Face Transformers, Neural Networks, NLP, Generative AI, Agentic AI
- Node.js, JavaScript, Pinecone, Vector Databases, OpenRouter, LLM Evaluation, Bitbucket Pipelines
- AWS, GCP, Kubernetes, Docker, LLM, GitHub Actions, SQL, CI/CD, NoSQL, Git, MLOps, PostgreSQL
- FastAPI, API Development, Automation Workflows, Cloud Platforms, GitHub, PyTorch, Postman
- N8N, Make.com, OpenClaw, Digital Ocean, MATLAB, OAuth 2.0, MLflow, AutoML, Process Automation
- React, HTML, CSS, Decision-Making, Power BI, Big Data Analytics, Model Evaluation, Data Analytics
- Railway, Bitbucket, Dropbox, OCR, Data Mining, Image Processing, Speech Recognition, Predictive Modeling, Data Visualization, Supabase

WORK EXPERIENCE

AI Engineer — KineticTech, Pakistan *Dec 2024 – Present*

- Serve as AI Engineer for Ballogy.com, developing an AI-powered coach assistant and personalized recommendation engine to guide player training and performance improvement.
- Own the data dictionary and data unification workstream: collect data from multiple heterogeneous source systems, map fields across sources, and standardize schemas to create a single unified data model.
- Lead backend development for the Ballogy.com analytics dashboard, building Python scripts and SQL queries to fetch and derive data from production databases, exposing the results through REST API endpoints consumed by a React frontend.
- Built Ballogy Model Lab, an AI model evaluation dashboard that benchmarks Ballogy's production AI against 80+ LLMs using automated multi-LLM judging and human feedback.
- Built a Python worker pipeline that syncs PostgreSQL data into Pinecone with deduplication and no external CDC tool, eliminating third-party costs.
- Developed a booking and fare prediction web application using React (frontend), Django (backend), and PostgreSQL (database), deployed via Railway App; implemented a Random Forest model ($R^2 \approx 0.87$) for fare estimation.
- Built a travel cost prediction platform covering data preprocessing, exploratory data analysis, and ML model training, integrated with FastAPI and a PostgreSQL backend, serving real-time predictions to 500+ active users per month, hosted on Digital Ocean.
- Designed a classification model to distinguish meditation from non-meditation audio samples, with Logistic Regression achieving the highest accuracy; processed .mat files by converting them into DataFrames for analysis and training.

AI Developer — Aug AI, Pakistan *Jan 2024 – Nov 2024*

- Built a travel cost prediction system integrating data preprocessing, EDA, ML model training, and FastAPI endpoints with a PostgreSQL backend, enabling real-time predictions for 500+ users per month.
- Designed and deployed an AI carbon emission calculator using XGBoost, integrated into the Yinn.ca platform and hosted on Digital Ocean with GitHub for version control.
- Built a Retrieval-Augmented Generation (RAG) system leveraging Llama 3.2, improving information retrieval and content generation efficiency by 40%.
- Created a CSV agent with LangChain enabling natural language queries on CSV data, reducing data analysis time by 60%.
- Trained an ensemble time series forecasting model (LSTM, Prophet, XGBoost, and LightGBM) achieving 96.7% accuracy for financial transaction predictions.

- Used Apify for data scraping from platforms such as LinkedIn and Indeed, automating the process through Zapier to streamline data collection workflows and improve lead acquisition by 50%.
- Engineered and trained a K-Means clustering model for data classification, visualizing outputs with 3D scatter plots to enhance data insights and decision-making.
- Planned and built an interactive sales dashboard using Power BI, enabling real-time data insights and improving decision-making efficiency by 45%.

AI Trainee — ICRL Lab, Pakistan ***Aug 2022 – Dec 2023***

- Contributed to detection and classification of labor equipment by training a YOLO model achieving 94%+ object detection and classification accuracy.
- Grew data annotation expertise, labeling data for effective model training and improving model performance by 25%.

IT Troubleshooter — Private Institute, Pakistan ***May 2018 – Jul 2022***

- Provided IT troubleshooting support, including software installations, Windows setup, and printer checks, improving system uptime by 60% and ensuring seamless operations.
- Diagnosed and resolved day-to-day hardware and software issues for staff, keeping computers and office systems running smoothly.

EDUCATION

Master of Science in IT — University of Education, Pakistan *Jul 2022*

Bachelor of Science in Computer Science — University of Punjab, Pakistan *Apr 2019*

PROJECTS

Ballogy Model Lab: AI Model Evaluation Dashboard — KineticTech ***Feb 2026 – Present***

- Built a full-stack dashboard that benchmarks Ballogy's production AI against 80+ LLMs on identical agent prompts and live data, using Node.js, JavaScript, Python, PostgreSQL, OpenRouter, Pinecone, Docker, and Bitbucket Pipelines.
- Scored anonymized, shuffled answers with multiple LLM judges plus automated checks, and produced win/tie/loss reports to guide model and prompt decisions.
- Added role-based access, audit logging, caching, request de-duplication, and a Dockerized CI/CD pipeline.

PostgreSQL-to-Pinecone Sync Pipeline — KineticTech ***Feb 2026 – Present***

- Built a Python worker pipeline with snapshots that syncs PostgreSQL data into Pinecone without CDC or external tools such as Estuary.
- Handled deduplication in Pinecone and eliminated third-party tooling costs.

Travel Cost Prediction Calculator — Aug AI ***Jan 2024 – Nov 2024***

- Developed and deployed a vehicle travel cost prediction system using Random Forest ($R^2 = 0.86$).
- Integrated the Google Places API for real-time distance calculations and built FastAPI endpoints with a PostgreSQL backend for real-time predictions on the client website.

AI Carbon Emission Calculator — Aug AI ***Jan 2024 – Nov 2024***

- Designed and implemented an AI-driven carbon emission calculator using XGBoost, integrated into the Yinn.ca platform.
- Deployed on Digital Ocean for scalability, with Git for version control.

Multi-Language Translation Model — Aug AI ***Jan 2024 – Nov 2024***

- Built a multi-language translation model leveraging Llama 3.2.
- Delivered high-quality, low-latency translations for cross-lingual communication.